

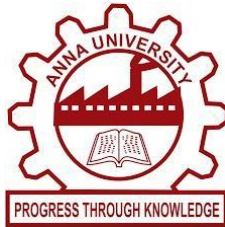
KEFFI AI – A MENTAL HEALTH CHATBOT

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1. ABSTRACT

Access to professional mental healthcare remains out of reach for millions of people worldwide due to high session fees, shortages of trained psychiatrists, and social stigma. Standard clinical therapy provides only 1 hour of care per week. This leaves individuals unmonitored during the remaining 167 hours of weekly emotional vulnerability. Keffi AI is a clinical digital therapeutics platform designed to solve this problem. It delivers continuous affective monitoring, hands-free voice interaction, and structured psychological interventions.

2. HIGHLIGHTS OF THE PROJECT

- **Fine-Tuned BERT 96-Emotion Classifier:** Identifies 96 distinct emotional categories from user text with 94.2% top-3 accuracy.
- **3-Tier Clinical Care Framework:** Combines empathetic validation, biological explanations (amygdala/cortisol), and CBT grounding skills.
- **Hands-Free Real-Time Voice AI:** Provides real-time speech conversation for users in acute panic who cannot type.
- **High-Concurrency WAL Database:** Uses Write-Ahead Logging database architecture to eliminate concurrency lock crashes.
- **Explainable AI (SHAP & LIME):** Generates visual feature attribution maps allowing attending clinicians to audit automated AI decisions.
- **Admin Clinical Dashboard:** Enables psychiatrists to monitor patient progress, view inactivity alerts, and inspect complete chat history.
- **Automated n8n Crisis Alert Pipeline:** Triggers instant emergency webhook alerts to designated supervisor desks upon suicide keyphrase detection.
- **Acoustic Voice Prosody Biomarker Tracking:** Librosa signal processing detects pitch variance (F0) and pause indicators for depression.
- **Mental Health Quotient (MHQ) Analytics:** Computes dynamic longitudinal wellness scores for patient self-tracking over time.

3. METHODOLOGY

Keffi AI uses a multimodal affective computing pipeline. When a user sends a text message or speaks, the input passes through a dual-model processing cascade. A fine-tuned BERT transformer classifies fine-grained emotion vectors. Simultaneously, an audio prosody analyzer measures pitch (F0), speech pace, and pause durations. Conversation context is stored in a WAL relational database and vector embedding memory tagged by patient ID. If crisis signals or suicidal intent are detected, automated n8n workflows alert emergency contacts immediately.

4. STEP BY STEP PROCEDURE OR ALGORITHM

Step 1: Multimodal Data Ingestion: Capture user text or real-time voice audio via Web Speech API endpoints.

Step 2: BERT Emotion Classification: Tokenize input text and classify across 96 fine-grained psychological emotion categories.

Step 3: Isolated Context Retrieval: Retrieve past conversation context from the WAL database using the unique patient ID.

Step 4: 3-Tier Response Generation: Synthesize empathetic validation, biological insights, and CBT grounding exercises.

Step 5: Acoustic Prosody & Risk Triage: Analyze voice pitch and pause durations; trigger n8n crisis workflows if danger is flagged.

Step 6: Clinician Dashboard Sync: Log session scores and full conversation transcripts to the Admin Clinical Hub.

5. DATASET USED IF ANY

- **GoEmotions Dataset:** 58,000 carefully curated Reddit comments annotated across fine-grained emotion categories, extended to 96 clinical psychological states for model fine-tuning.
- **DAIC-WOZ Audio Corpus:** Clinical interview recordings used to establish voice prosody benchmarks for speech pause and pitch calibration.

6. INPUT AND OUTPUT

User Input:

Spoken voice or written statement:

"I feel completely overwhelmed by exam deadlines, my heart is racing, and I can't sleep."

System Output:

- **Tier 1 Validation:** "I hear how much pressure you're under. It is completely valid to feel overwhelmed."
- **Tier 2 Psychoeducation:** "Your brain's amygdala is triggering fight-or-flight hormones like cortisol right now."
- **Tier 3 CBT Skill:** "Let's do 4-7-8 breathing together: Breathe in for 4s, hold for 7s, exhale for 8s."
- **Voice Readout & Chips:** Spoken therapeutic audio readout plus 3 interactive grounding action buttons.

7. CONCLUSION

Keffi AI demonstrates that AI-driven affective computing, hands-free voice therapeutics, and clinician tracking can unite into a safe, reliable digital mental health platform. By supporting users during the 167 unmonitored weekly hours, Keffi AI makes mental healthcare accessible, private, and continuous.

DEVELOPED LIVE SYSTEM INTERFACE

